SkyCoin444 v10 Live Platform - System Design & Architecture

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Executive Summary

SkyCoin444 v10 is a comprehensive Web3 fintech super-app that combines day trading, AI-powered market analysis, social engagement, and gamified user progression into a unified platform. The system is built on a modern tech stack featuring React 19, TypeScript, tRPC, Express, and MySQL, with real-time data simulation and LLM-powered intelligence.

The platform enables users to execute simulated trades, receive AI-driven market insights, engage in a social feed with tip-based rewards, and track their performance across multiple leaderboards. All data is persisted in a relational database with comprehensive schema design supporting trading, social interactions, vault management, and user progression.

System Architecture

Technology Stack

The SkyCoin444 v10 platform is built on a modern, production-ready technology foundation:

Component	Technology	Version	Purpose
Frontend Framework	React	19.2.1	UI rendering with hooks and concurrent features
Type Safety	TypeScript	5.9.3	End-to-end type safety across client and server
Styling	Tailwind CSS	4.1.14	Utility-first CSS with dark theme support
Backend Framework	Express	4.21.2	HTTP server and API routing
RPC Framework	tRPC	11.6.0	Type-safe remote procedure calls
Database	MySQL	8.0+	Relational data persistence
ORM	Drizzle	0.44.5	Type-safe database queries
Data Visualization	Recharts	2.15.2	React-based charting library
UI Components	shadcn/ui	Latest	Accessible, customizable components
Bundler	Vite	7.1.7	Fast build tool and dev server
Package Manager	pnpm	10.4.1	Efficient monorepo management

Architecture Diagram

The system follows a three-tier architecture with clear separation of concerns:

Frontend Layer (Client): React-based SPA with Tailwind CSS styling, handling all user interactions, real-time updates, and data visualization.

API Layer (tRPC): Type-safe RPC procedures bridging frontend and backend, with automatic serialization and error handling.

Backend Layer (Server): Express-based HTTP server managing authentication, database operations, LLM integration, and real-time simulations.

Data Layer (Database): MySQL with Drizzle ORM providing schema-first database design and type-safe queries.

Database Schema

The platform uses a comprehensive 12-table schema designed to support all features:

Core Tables

users - Stores user identity and authentication state with OAuth integration. Fields include `id`, `openId`, `name`, `email`, `loginMethod`, `role`, `createdAt`, `updatedAt`, `lastSignedIn`.

portfolios - Tracks user portfolio metrics including total value, XP, and level. One-to-one relationship with users table.

holdings - Maintains user asset positions with asset symbol and amount. Supports multiple holdings per user.

Trading Tables

trades - Records all executed trades with pair, type (buy/sell), amount, price, total, and status. Enables order history and analytics.

Social Tables

posts - Stores social feed content with engagement metrics (likes, replies, shares, tips), AI ranking score, and sentiment analysis.

messages - Direct messaging between users with tip-in-chat functionality and read status tracking.

chatHistory - Persists AI copilot conversation history with role (user/assistant) and content.

Vault & Staking Tables

vaults - Time-locked positions with tier selection (flex, quarter, annual, diamond), APY tracking, and unlock dates.

Gamification Tables

leaderboard - Tracks user rankings across five categories: XP, mining, staking, trading, and referrals.

apiKeys - Manages user API keys with scopes, creation date, and last used timestamp.

referrals - Tracks referral relationships with tier progression and earned rewards.

onboardingProgress - Monitors user onboarding completion with step tracking and XP rewards.

Feature Specifications

1. Live Day Trading Dashboard

The trading dashboard provides a professional-grade interface for executing simulated trades with real-time market data.

Key Components:

- **Price Chart:** Real-time candlestick/line chart using Recharts with 24-hour historical data
- **Order Book:** Live bid/ask orders with price levels and volume
- **Order Execution:** Buy/sell form with amount, price, and order type selection
- **Order History:** Recent trades with timestamp, pair, type, amount, and price
- **Market Stats:** 24h high, low, volume, and current price display

Real-Time Simulation: Prices update every second using a random walk algorithm that maintains realistic volatility. The simulation engine generates new price points based on the previous price with a small random change (±0.05% per second).

Data Persistence: All trades are recorded in the `trades` table with status tracking (pending, filled, cancelled). Order history is queryable via tRPC procedure `trading.getOrders`.

2. Rogue AI Trading Copilot

The AI copilot provides intelligent market analysis and trade suggestions powered by LLM integration.

Capabilities:

- **Market Analysis:** Real-time sentiment scoring based on current price, volume, and historical trends
- **Trade Suggestions:** AI-generated buy/sell recommendations with reasoning
- **Chat Interface:** Conversational interface for asking questions about market conditions
- **Context Awareness:** Copilot understands user's current portfolio and positions
- **Streaming Responses:** Real-time message streaming for responsive user experience

LLM Integration: The copilot uses the Manus built-in LLM service via `invokeLLM` helper function. System prompt instructs the model to act as a financial advisor with knowledge of cryptocurrency markets.

Chat History: All conversations are persisted in the `chatHistory` table with role (user/assistant) and timestamp for context continuity.

3. ShadowChat Social Feed

The social feed enables users to share market insights, engage with other traders, and earn rewards through tipping.

Features:

- **Post Creation:** Text and image posts with markdown support
- **Engagement Metrics:** Likes, replies, shares, and tips tracking
- **AI Ranking:** Posts are ranked using an AI algorithm that considers engagement, recency, and sentiment
- **Tipping System:** Users can tip posts in-app currency to reward valuable insights
- **Infinite Scroll:** Pagination-based feed loading with lazy loading
- **User Profiles:** Post author information with profile links

AI Ranking Algorithm: Posts are ranked using a composite score: `aiRank = (likes * 2) + (tips * 5) + (shares * 3) - (age_in_hours * 0.1)`. This encourages quality engagement while gradually deprioritizing older posts.

Data Model: Posts are stored in the `posts` table with fields for content, images, engagement metrics, AI rank, and sentiment analysis.

4. Analytics Dashboard

The analytics dashboard provides comprehensive portfolio and market insights using interactive charts.

Visualizations:

- **Portfolio Value Chart:** Line chart showing portfolio value over time (24h, 7d, 30d, 90d, 1y)
- **Asset Allocation:** Pie chart showing percentage breakdown of holdings
- **Holdings Table:** Detailed breakdown of each asset with amount, value, and percentage
- **Performance Metrics:** Cards displaying total value, daily change %, and top performers
- **TVL & Volume:** Key market metrics displayed as cards

Data Sources: Portfolio data is calculated from the `holdings` table combined with simulated price data. Historical data is generated by the simulation engine.

Interactive Features: Time range selector allows users to view different time periods. Charts update in real-time as prices change.

API Procedures (tRPC)

All backend functionality is exposed through type-safe tRPC procedures:

Trading Procedures

- `trading.createOrder` - Execute a new trade (buy/sell)

- `trading.getOrders` - Retrieve user's trade history

Portfolio Procedures

- `portfolio.get` - Fetch user's portfolio data

Social Procedures

- `social.createPost` - Create a new social post
- `social.getFeed` - Retrieve paginated social feed

Messaging Procedures

- `messages.send` - Send a direct message with optional tip

Vault Procedures

- `vault.create` - Create a new time-locked vault position
- `vault.getVaults` - Retrieve user's vault positions

Leaderboard Procedures

- `leaderboard.get` - Retrieve rankings for a specific category

Authentication Procedures

- `auth.me` - Get current user information
- `auth.logout` - Clear session and logout

Real-Time Features

Price Simulation Engine

The platform includes a built-in price simulation engine that generates realistic market data without requiring external APIs. The engine:

- Generates prices using a random walk model with mean reversion
- Updates prices every second with configurable volatility
- Maintains order book with realistic bid/ask spreads
- Persists trade history to database

WebSocket Integration (Future)

The current implementation uses polling via React hooks. Future versions will implement WebSocket for true real-time updates with lower latency.

Security & Authentication

OAuth Integration

The platform uses Manus OAuth for secure user authentication. The flow:

1. User clicks login, redirected to OAuth portal
2. After authentication, callback handler creates/updates user session
3. Session cookie is set with secure, HttpOnly, SameSite flags
4. Protected procedures check session context before execution

Protected Procedures

All user-specific operations use `protectedProcedure` which validates the session context. Attempting to access protected procedures without authentication returns a 401 error.

API Key Management

Users can generate API keys with granular scopes (read, trade, withdraw, admin). Keys are stored with hashed secrets and can be revoked at any time.

Deployment & Hosting

The platform is deployed on Manus Cloud with the following characteristics:

- **Auto-scaling:** Handles variable traffic with automatic instance scaling
- **Database:** Managed MySQL instance with automatic backups
- **Storage:** S3-compatible object storage for user uploads and assets
- **CDN:** Global CDN for static assets and API responses
- **Monitoring:** Built-in analytics and error tracking

Environment Variables

Key environment variables configured at deployment:

- `DATABASE_URL` - MySQL connection string
- `JWT_SECRET` - Session signing secret
- `VITE_APP_ID` - OAuth application ID
- `OAUTH_SERVER_URL` - OAuth provider URL
- `BUILT_IN_FORGE_API_KEY` - LLM service key

Performance Considerations

Frontend Optimization

- **Code Splitting:** Routes are lazy-loaded to reduce initial bundle size
- **Image Optimization:** User uploads are processed and cached
- **Caching:** tRPC queries are cached with React Query
- **Memoization:** Components use React.memo to prevent unnecessary re-renders

Backend Optimization

- **Database Indexing:** Key fields (userId, createdAt) are indexed

- **Query Optimization:** Drizzle ORM generates efficient SQL
- **Connection Pooling:** MySQL connection pool prevents resource exhaustion
- **Rate Limiting:** API endpoints are rate-limited per user

Real-Time Performance

- **Simulation Efficiency:** Price updates are calculated client-side to reduce server load
- **Batch Updates:** Multiple price updates are batched before UI re-render
- **Debouncing:** Chart updates are debounced to 1 second intervals

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Future Roadmap

Phase 2 Features

- **WebSocket Integration:** Real-time price updates via WebSocket
- **Advanced Analytics:** Machine learning-based price predictions
- **Mobile App:** React Native mobile version
- **Payment Integration:** Stripe for real fund deposits

Phase 3 Features

- **Cross-Chain Bridge:** Support for multiple blockchain networks
- **Automated Trading:** Bot-based trading strategies
- **Advanced Charting:** TradingView-like advanced charts
- **Community Features:** User groups and collaborative trading

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Conclusion

SkyCoin444 v10 represents a comprehensive, production-ready fintech platform that combines trading, AI intelligence, and social engagement. The architecture is scalable, secure, and built on proven technologies. The modular design allows for easy feature additions and maintenance.

The platform successfully demonstrates:

- Real-time data simulation and visualization
- LLM-powered AI features
- Social-Fi integration with gamification
- Type-safe full-stack development
- Professional fintech UX/UI

References

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